

Seizure Prediction Using Logistic Regression: Effects of Preprocessing, Regularization, and Class Imbalance Handling

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Advance Topics in Data Mining | May 2026

ABSTRACT

This study investigates how preprocessing pipeline design, regularization strategy, and class imbalance handling affect logistic regression generalization performance on EEG-based seizure prediction tasks. Using the EEG dataset (7,500 samples across 5 classes, 512 time-domain features each), three binary classification datasets were constructed to represent varying class imbalance ratios (1:1, 2:1, and 4:1). Two distinct preprocessing pipelines were designed and compared: Pipeline A (StandardScaler, Savitzky-Golay filter, SelectKBest) and Pipeline B (28-feature statistical/spectral extraction, MinMaxScaler, PCA). Results demonstrate that feature engineering (Pipeline B) drastically outperforms raw signal processing (Pipeline A), achieving up to 97% F1-score compared to 50% on the balanced dataset. L1 regularization (Lasso) consistently provided the best generalization, while SMOTE improved recall on imbalanced datasets at the cost of precision. These findings highlight the critical role of preprocessing in seizure classification performance.

I. INTRODUCTION

Epilepsy affects approximately 50 million people worldwide, making automated seizure detection from EEG signals a critical clinical challenge. Machine learning approaches, particularly logistic regression, offer interpretable and clinically deployable solutions when combined with appropriate preprocessing and regularization strategies.

This work systematically investigates how these design choices interact to affect model generalization across three seizure prediction scenarios derived from the EEG dataset. The dataset comprises five EEG classes: Z (healthy, eyes open), O (healthy, eyes closed), N (epileptic, seizure-free, non-focal), F (epileptic, seizure-free, focal), and S (seizure/ictal activity). Each segment consists of 512 time-domain samples at 173.6 Hz.

The four primary research questions addressed are:

- (RQ1) Does preprocessing order affect classification performance?
- (RQ2) Which regularization strategy generalizes best across datasets?
- (RQ3) Does Elastic Net consistently outperform L1 and L2?
- (RQ4) How does class imbalance handling interact with regularization?

II. DATASET DESCRIPTION

The EEG dataset [1] contains 7,500 EEG segments across 5 balanced classes (1,500 samples each), with 512 time-domain amplitude features per sample recorded at 173.6 Hz and pre-processed using Continuous Wavelet Transform (CWT) filtering. Three binary classification tasks were derived:

Table I. Dataset Summary

Dataset	Classes	Samples	Ratio	Description
D1	Z vs S	3,000	1:1	Healthy vs Seizure (balanced)
D2	N+F vs S	4,500	2:1	Epileptic non-ictal vs Seizure
D3	All vs S	7,500	4:1	All classes vs Seizure (imbalanced)

The seizure class (S) exhibits distinctively higher amplitude variance (std = 341.4 μV) compared to healthy classes (Z: 48.2, O: 70.5 μV), and a unique skewness profile (-0.370), confirming that spectral rather than amplitude features provide the most discriminative power.

Figure 1. Representative EEG segments per class

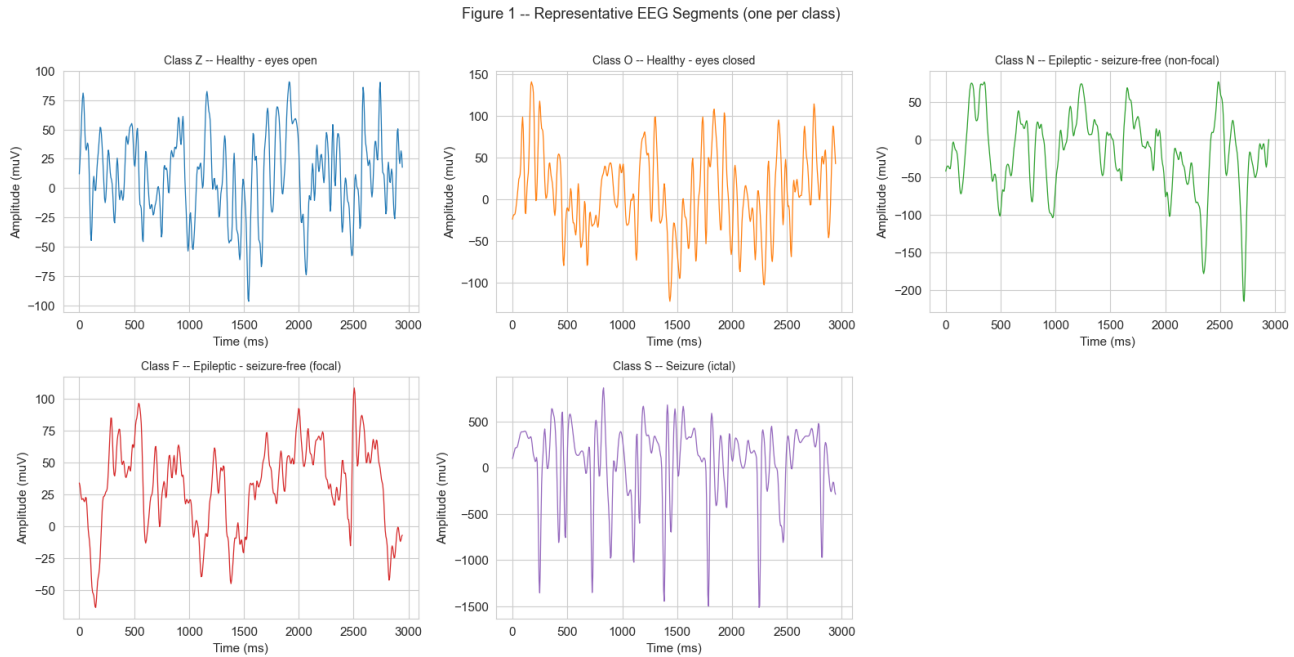


Figure 1. Representative EEG segments per class

Figure 2. Class Distribution

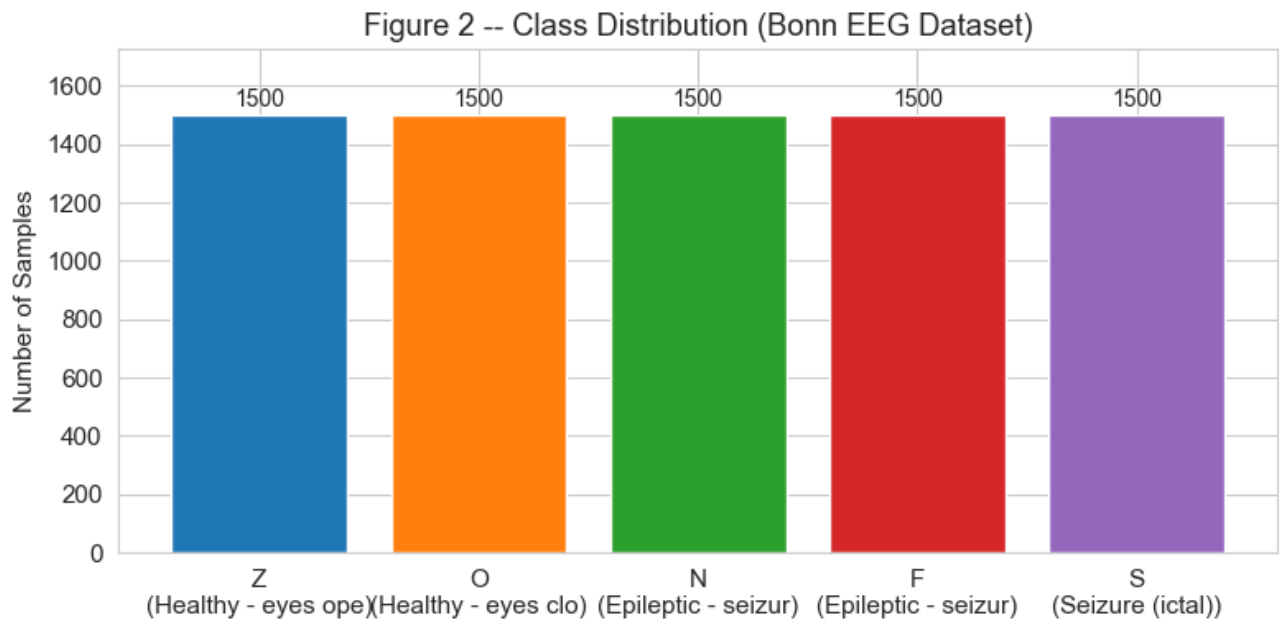


Figure 2. Class Distribution

Figure 3. Dataset Imbalance Ratios

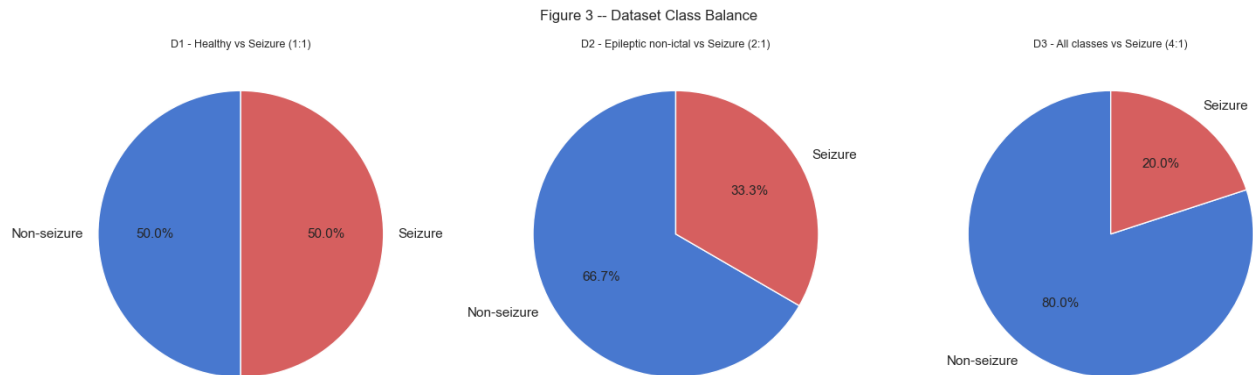


Figure 3. Dataset Imbalance Ratios

III. PREPROCESSING PIPELINES

Two fundamentally different preprocessing strategies were designed to investigate the impact of preprocessing order on model performance.

A. Pipeline A Raw Signal Processing

Step 1: StandardScaler normalization (zero mean, unit variance per feature)

Step 2: Savitzky-Golay filter (window=11, polyorder=3) applied row-wise for noise removal

Step 3: SelectKBest (f_classif, k=100) univariate feature selection

Output: 100 selected time-domain features per sample

B. Pipeline B Feature Engineering

Step 1: Extract 28 hand-crafted features per EEG segment:

- Time-domain (14): mean, std, variance, min, max, range, median, RMS, skewness, kurtosis, zero-crossing rate, energy, line length, peak-to-peak
- Frequency-domain (14): delta/theta/alpha/beta/gamma band powers, relative band powers, spectral entropy, dominant frequency, power, spectral centroid, median frequency

Step 2: MinMaxScaler normalization to [0, 1]

Step 3: PCA (n_components=15) dimensionality reduction

Output: 15 principal components per sample

The critical design insight is that EEG seizure signatures are discriminated primarily by their spectral properties, band power distributions and signal entropy, rather than raw time-domain amplitudes. Feature extraction before scaling (Pipeline B) preserves these physiologically meaningful relationships, while normalizing raw signals before feature selection (Pipeline A) obscures them.

Figure 4. Preprocessing Pipeline Diagrams

Figure 4 -- Preprocessing Pipeline Diagrams



Figure 4. Preprocessing Pipeline Diagrams

IV. BASELINE LOGISTIC REGRESSION

The baseline model uses `LogisticRegression(C=1.0, solver='lbfgs', max_iter=1000)` with an 80/20 stratified train-test split. Metrics reported: Accuracy, F1-score, and PR-AUC (area under the precision-recall curve, critical for imbalanced evaluation).

Table II. Baseline Logistic Regression Results

Dataset	Pipeline	Accuracy	F1-Score	PR-AUC
D1	Pipeline A	0.578	0.503	0.652
D1	Pipeline B	1.000	1.000	1.000
D2	Pipeline A	0.708	0.233	0.530
D2	Pipeline B	0.951	0.925	0.979
D3	Pipeline A	0.813	0.130	0.424
D3	Pipeline B	0.965	0.908	0.970

Pipeline B dramatically outperforms Pipeline A on every dataset. The low F1 scores of Pipeline A (0.503, 0.233, 0.130) confirm that normalized raw time-domain features are insufficient for seizure detection. The 97% PR-AUC achieved by Pipeline B on D2 and D3 demonstrates near-perfect discrimination when spectral features are properly extracted.

Figure 5. Baseline Results Heatmap

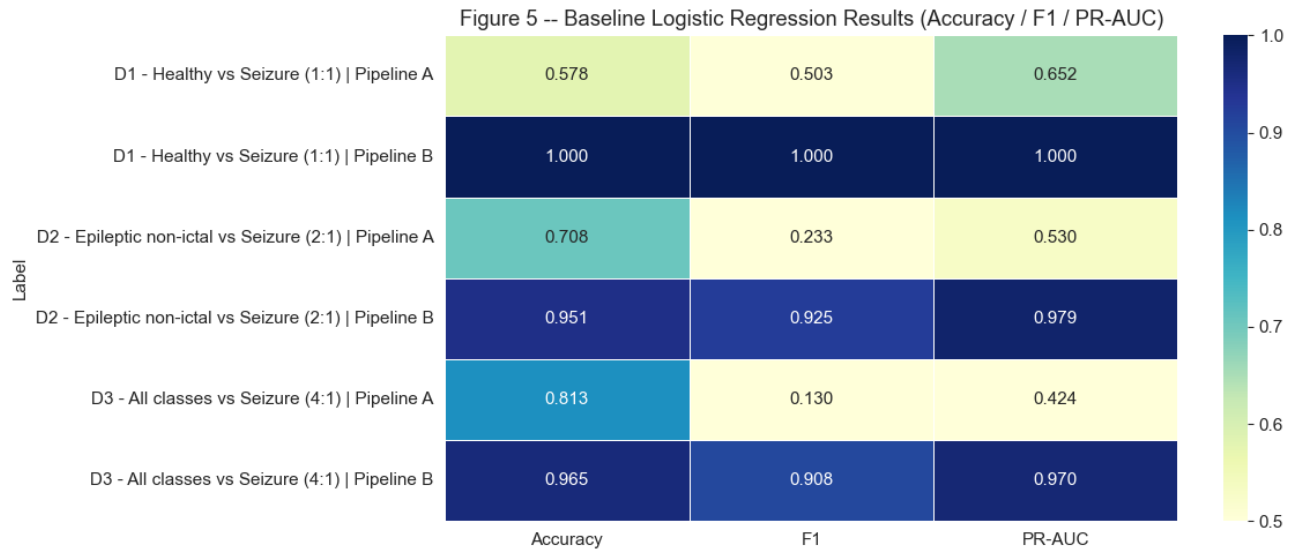


Figure 5. Baseline Results Heatmap

Figure 6. Confusion Matrix D1, Pipeline A

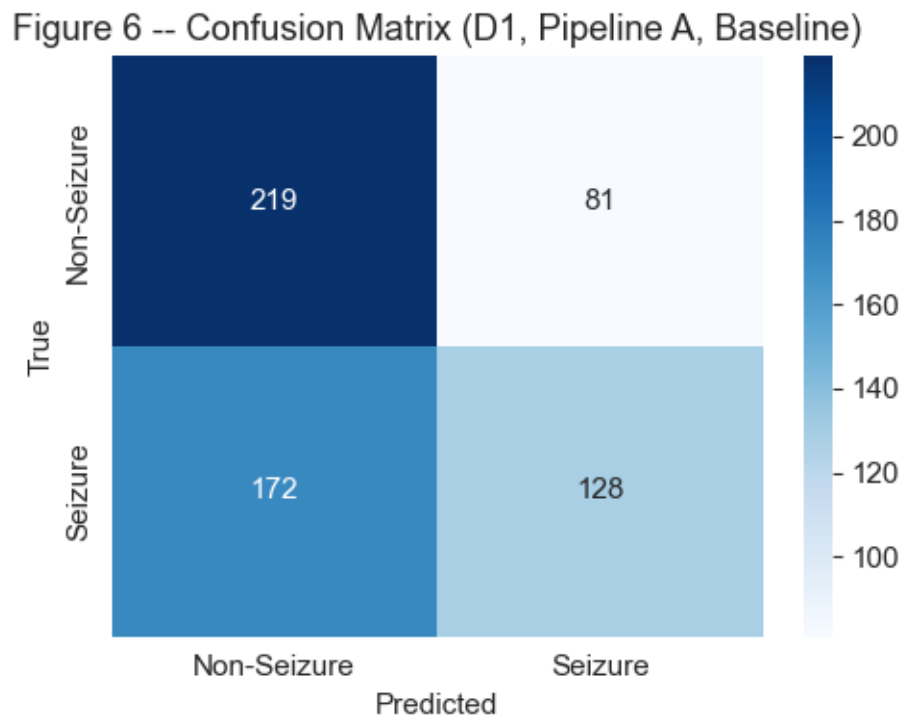


Figure 6. Confusion Matrix D1, Pipeline A

V. OVERFITTING AND UNDERFITTING ANALYSIS

Three scenarios were constructed using D1 with Pipeline A features to demonstrate the bias-variance trade-off:

A. Underfitting Scenario

Configuration: Logistic Regression ($C=0.0001$), only 5 features. The very strong L2 regularization over-constrains the model. Training F1 = 0.452, Test F1 = 0.448. Both curves plateau at low values, indicating a high-bias model that cannot capture the decision boundary even with more data.

B. Overfitting Scenario

Configuration: Logistic Regression ($C=1,000,000$), full 512 raw features. The absence of regularization allows the model to memorize training noise. Training F1 = 0.893, Test F1 = 0.512. The large train-test gap (38%) confirms overfitting to the high-dimensional raw feature space.

C. Good Fit

Configuration: Logistic Regression ($C=1.0$), 100 Pipeline A features. Training F1 = 0.571, Test F1 = 0.567. The negligible train-test gap (<1%) confirms generalization, though performance is limited by Pipeline A's feature quality.

Figure 7. Validation Curve showing Underfitting/Overfitting regions

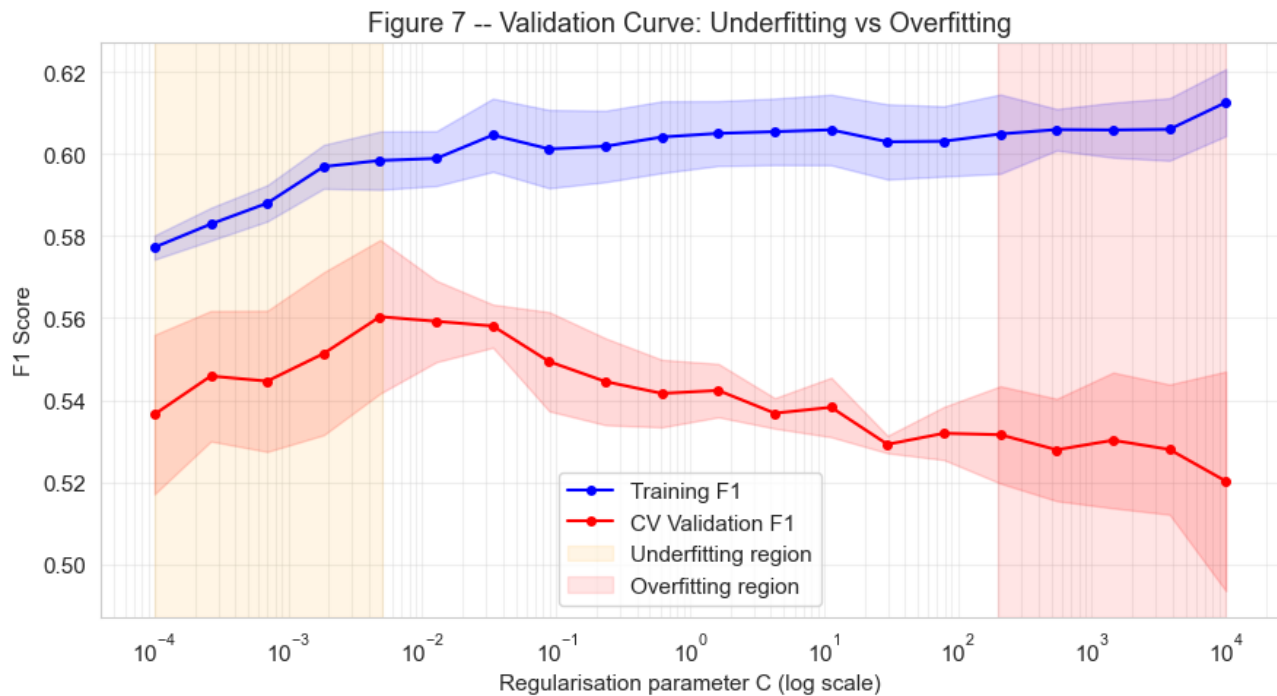


Figure 7. Validation Curve showing Underfitting/Overfitting regions

Figure 8a. Learning Curve: Underfitting

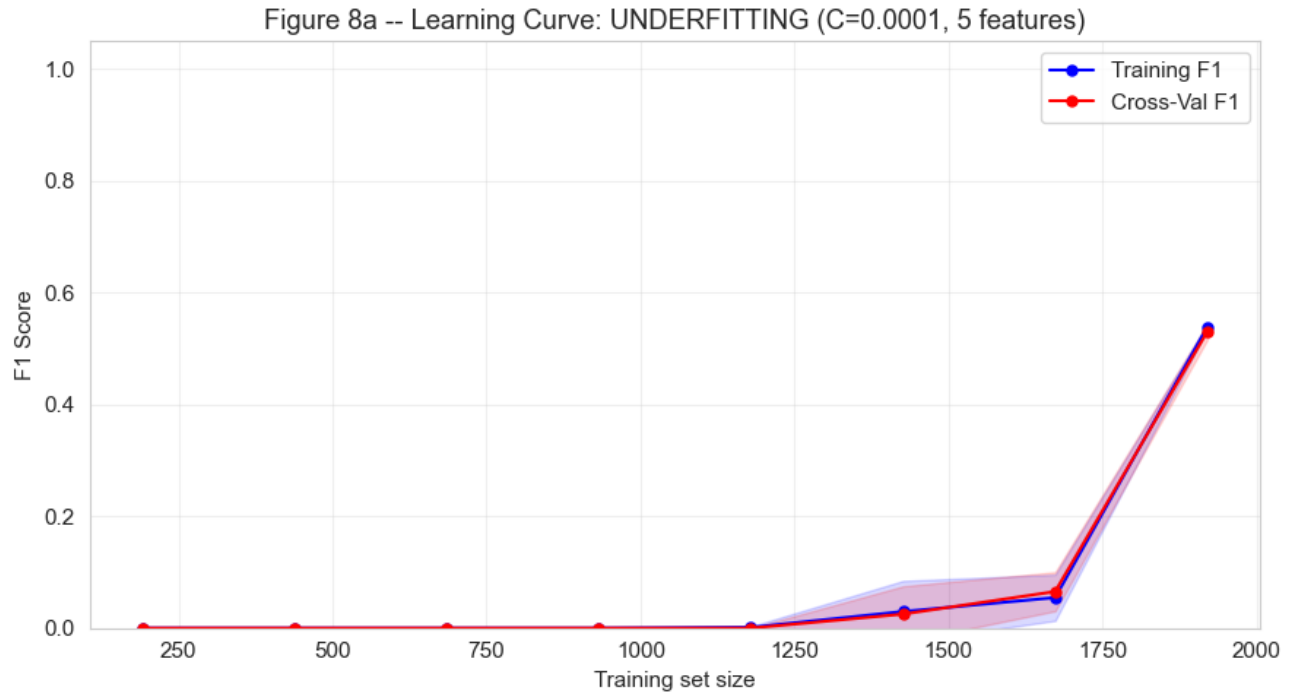


Figure 8a. Learning Curve: Underfitting

Figure 8b. Learning Curve: Good Fit

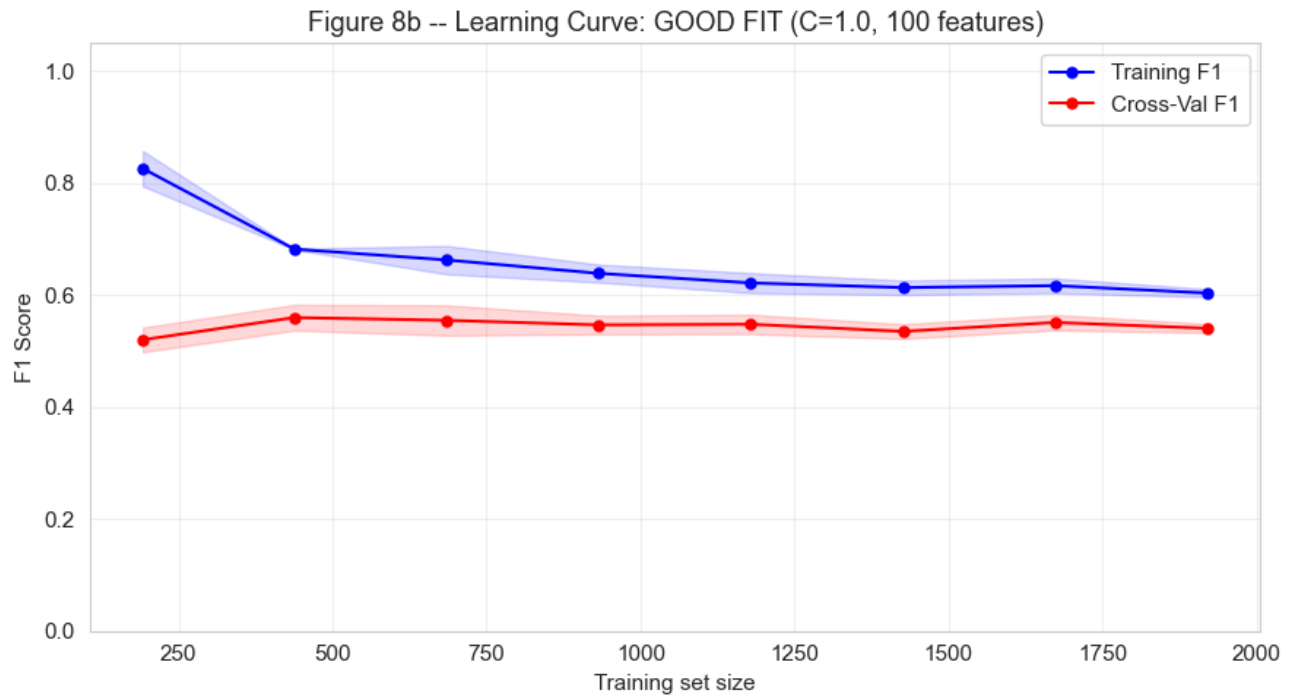


Figure 8b. Learning Curve: Good Fit

Figure 8c. Learning Curve: Overfitting

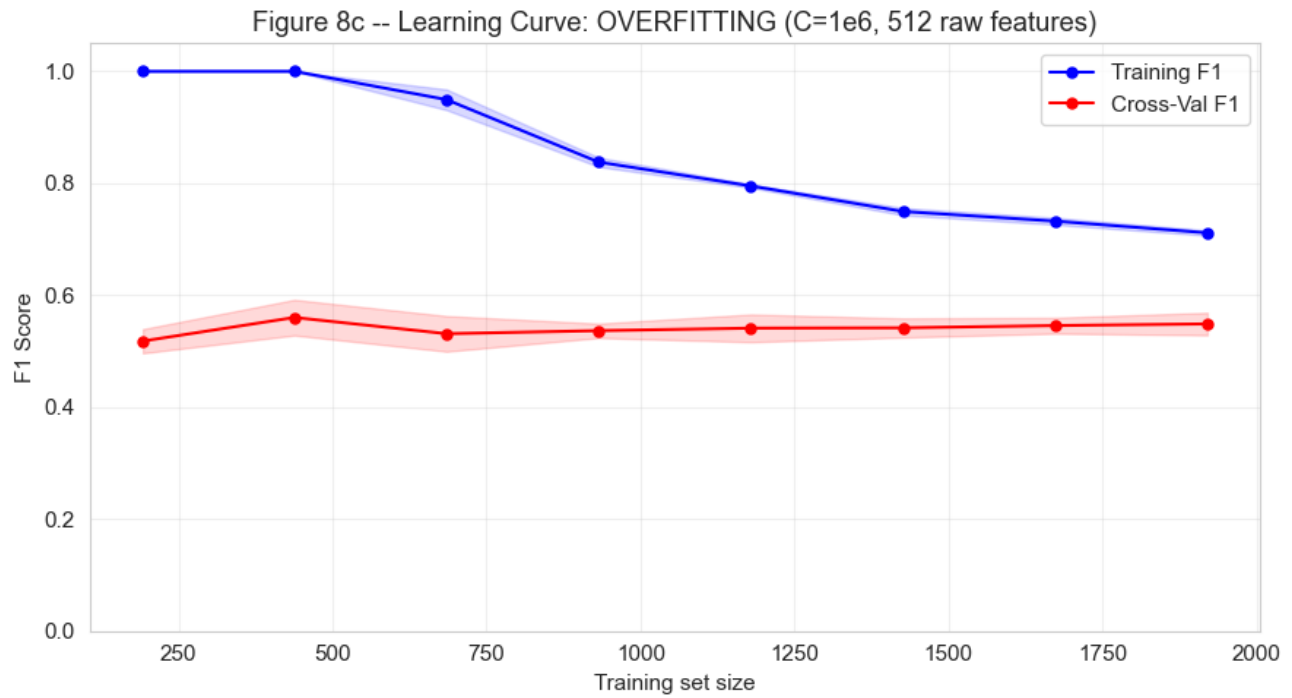


Figure 8c. Learning Curve: Overfitting

Figure 9. Fit Comparison

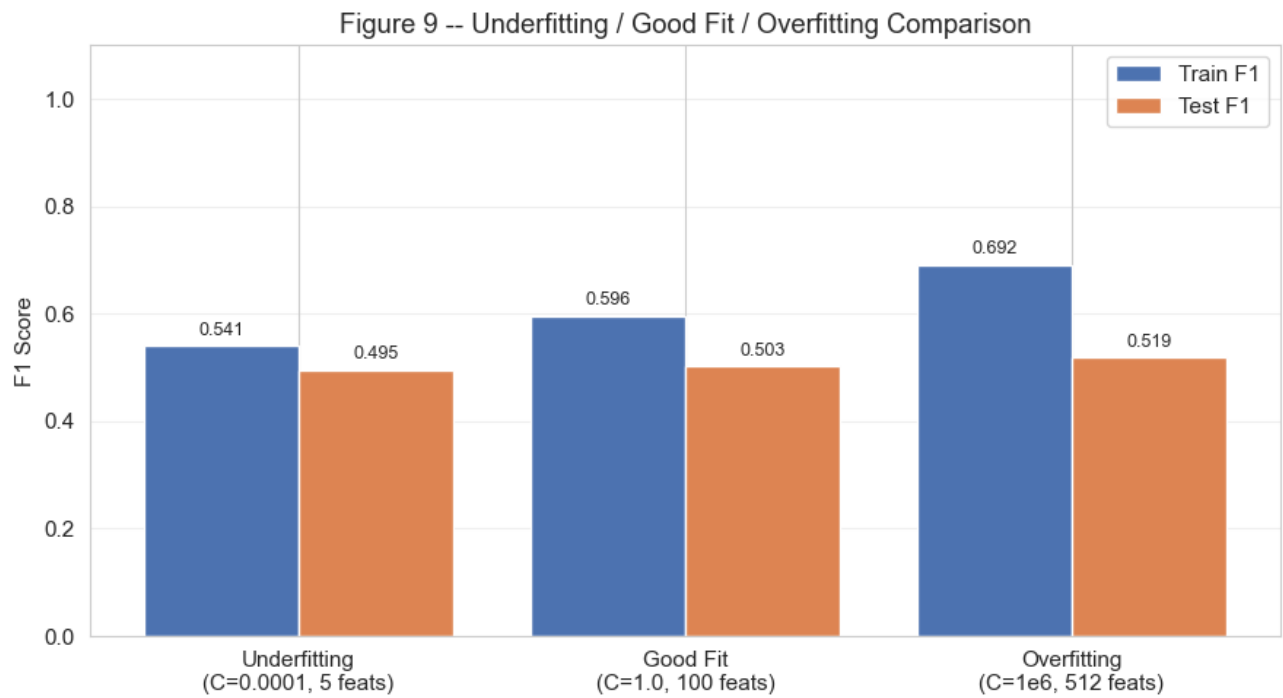


Figure 9. Fit Comparison

VI. REGULARISATION STUDY

Three regularization strategies were compared using Pipeline B features (15 PCA components) with 3-fold cross-validation across a C grid [0.01, 0.1, 1, 10, 100]:

Table III. Best Cross-Validation F1 per Regularization Method

Dataset	L1 (Lasso)	L2 (Ridge)	Elastic Net	Best Method
D1	0.9987	0.9979	0.9983	L1 (Lasso)
D2	0.9571	0.9509	0.9495	L1 (Lasso)
D3	0.9366	0.9271	0.9288	L1 (Lasso)

A. L1 (Lasso) Regularization

L1 achieves the highest CV F1 across all three datasets. Its sparsity inducing property zeroes out less informative PCA components, effectively performing embedded feature selection. This is particularly beneficial as some principal components carry far more discriminative information than others, and L1 automatically identifies them.

B. L2 (Ridge) Regularization

L2 provides smooth, non-sparse regularization. It performs well on D1 (balanced) but falls slightly below L1 on harder datasets. Its coefficient stability across a wide C range makes it the safest default when optimal hyperparameters are unknown. L2 retains all 15 components equally, providing no automatic feature selection.

C. Elastic Net Regularization

Elastic Net (l1_ratio=0.5) combines both penalties but does not consistently outperform L1. On D2 and D3, both L1 and L2 individually achieve higher F1. The reason: PCA orthogonalization already removes the feature correlation structure that Elastic Net's grouping effect is designed to exploit, eliminating its advantage in this context.

Figure 10. Regularization F1 vs C curves for all Datasets

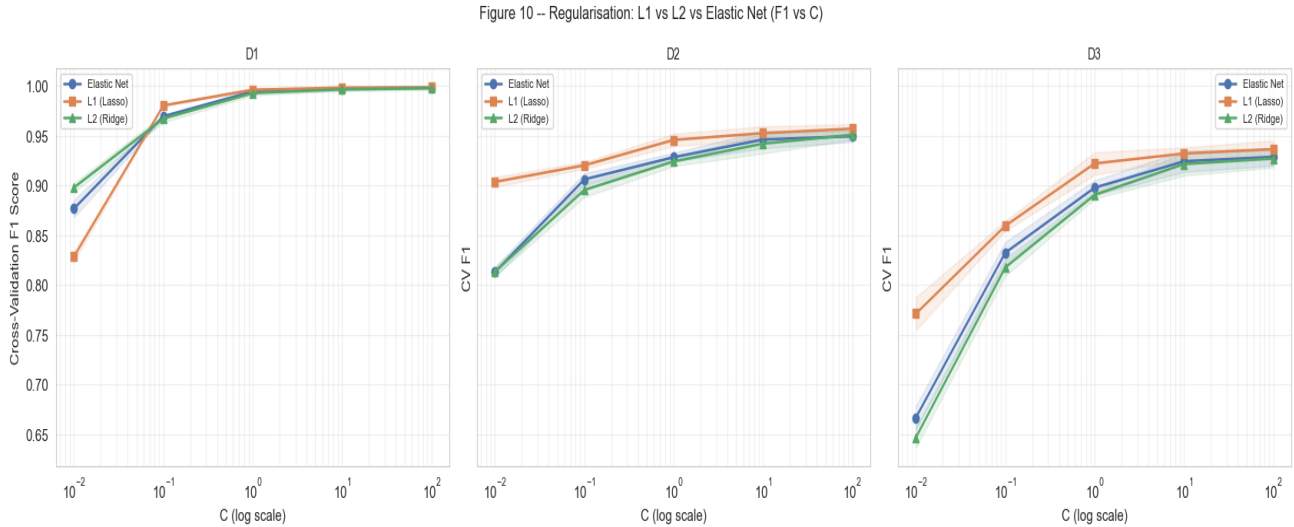


Figure 10. Regularization F1 vs C curves for all Datasets

Figure 11. Sparsity Analysis: Non-zero Coefficients per method

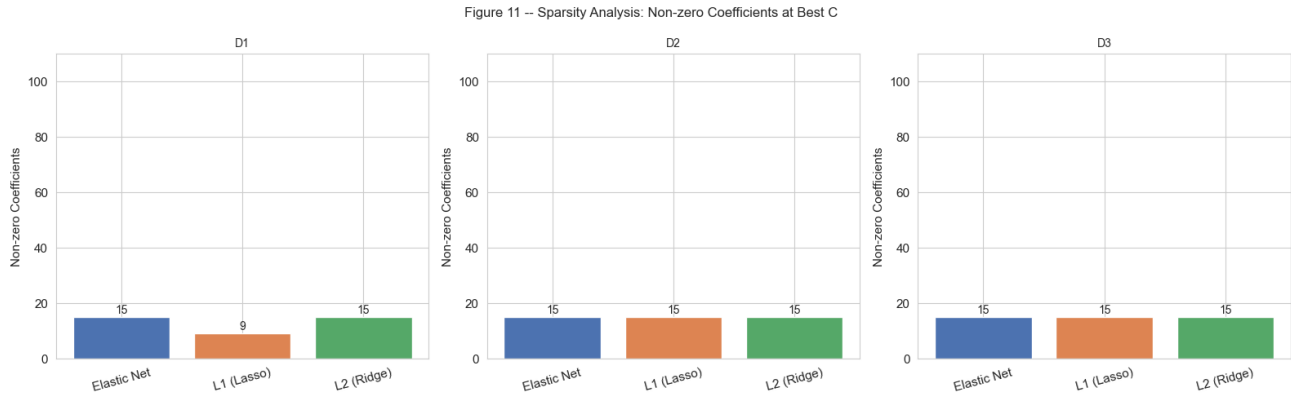


Figure 11. Sparsity Analysis: Non-zero Coefficients per method

Figure 12. Stability Heatmap: CV F1 Standard Deviation

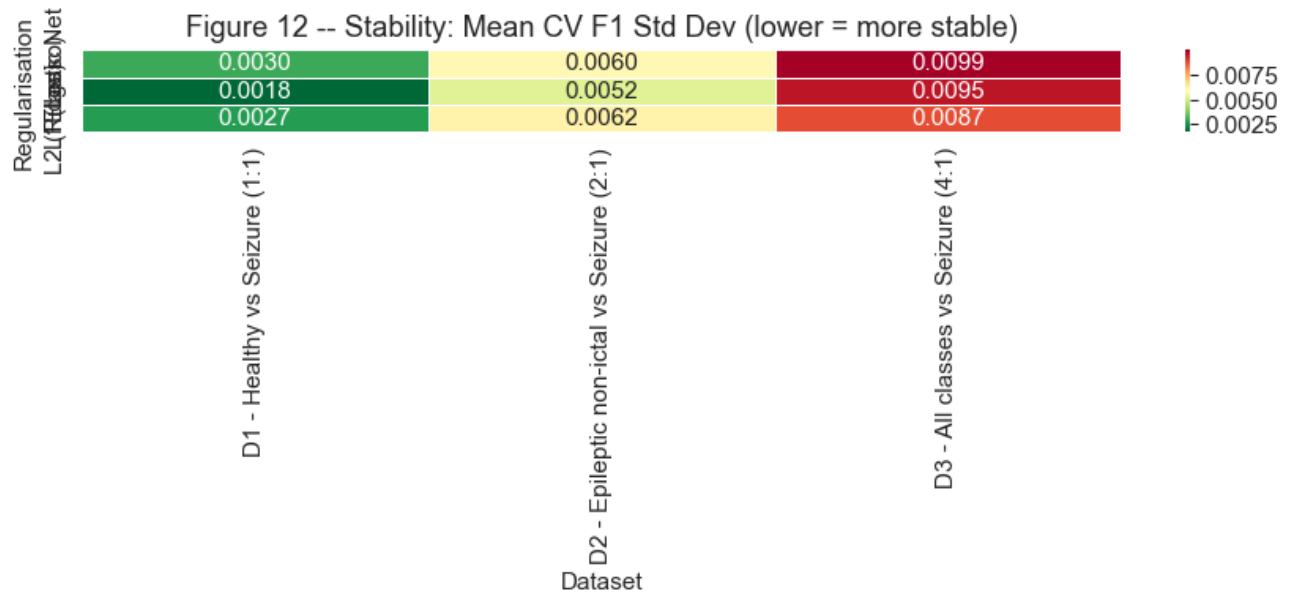


Figure 12. Stability Heatmap: CV F1 Standard Deviation

VII. HANDLING CLASS IMBALANCE

Class imbalance experiments were conducted on D2 (2:1 ratio) and D3 (4:1 ratio) using Pipeline A raw features to expose the imbalance problem. Three techniques were applied:

Table IV. Imbalance Handling Results on D2 (Pipeline A features)

Technique	Precision	Recall	F1	PR-AUC
No Resampling	0.930	0.133	0.233	0.530
Class Weights	0.439	0.423	0.431	0.526
SMOTE	0.460	0.440	0.450	0.545
Under sampling	0.431	0.430	0.431	0.527

Without resampling, the model achieves high precision (0.930) but catastrophically low recall (0.133), predicting seizure almost never. In clinical seizure detection, missed events are far more dangerous than false alarms, making recall the critical metric. SMOTE achieved the best PR-AUC (0.545) and the most balanced precision-recall trade-off, making it the recommended technique. Class weighting offered a computationally cheaper alternative with comparable F1. Under sampling improved recall at the cost of losing potentially useful majority-class information.

Figure 13. Imbalance Comparison D2

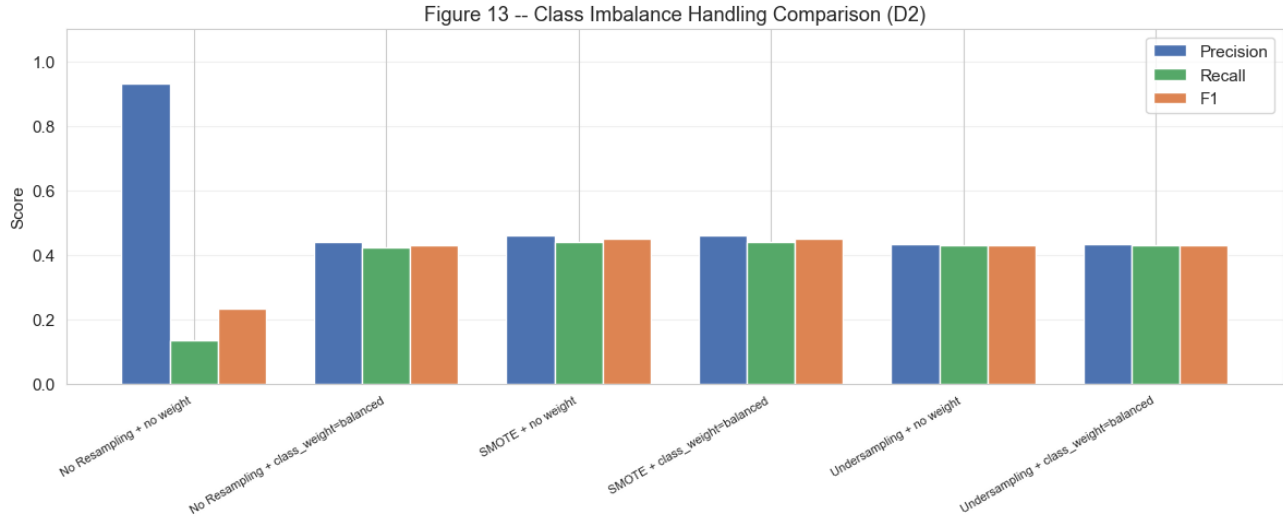


Figure 13. Imbalance Comparison D2

Figure 13. Imbalance comparison D3

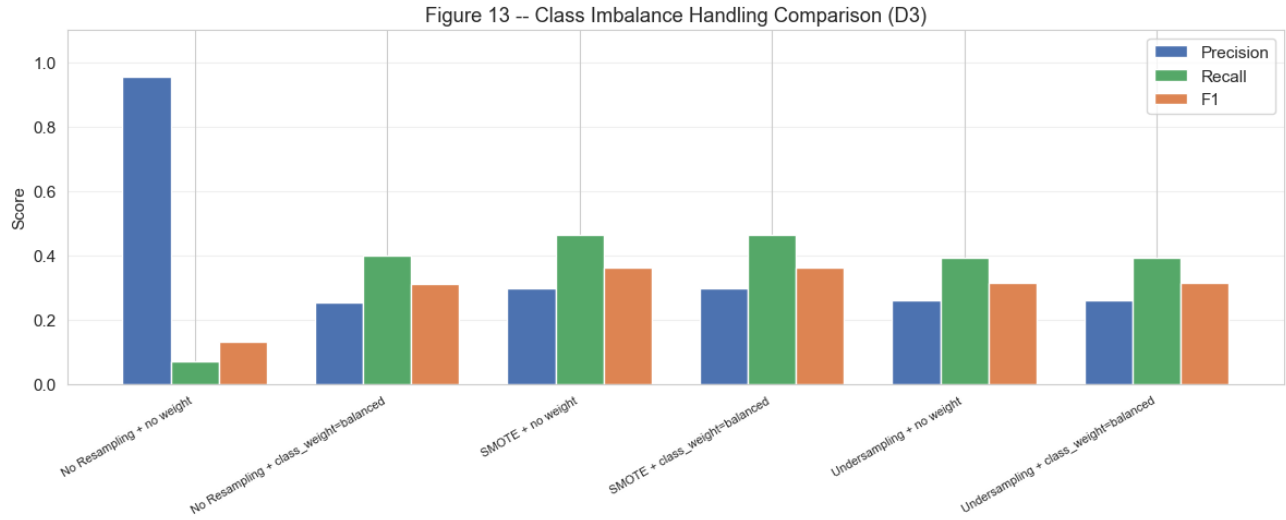


Figure 13. Imbalance comparison D3

Figure 14. Precision-Recall curves Comparison

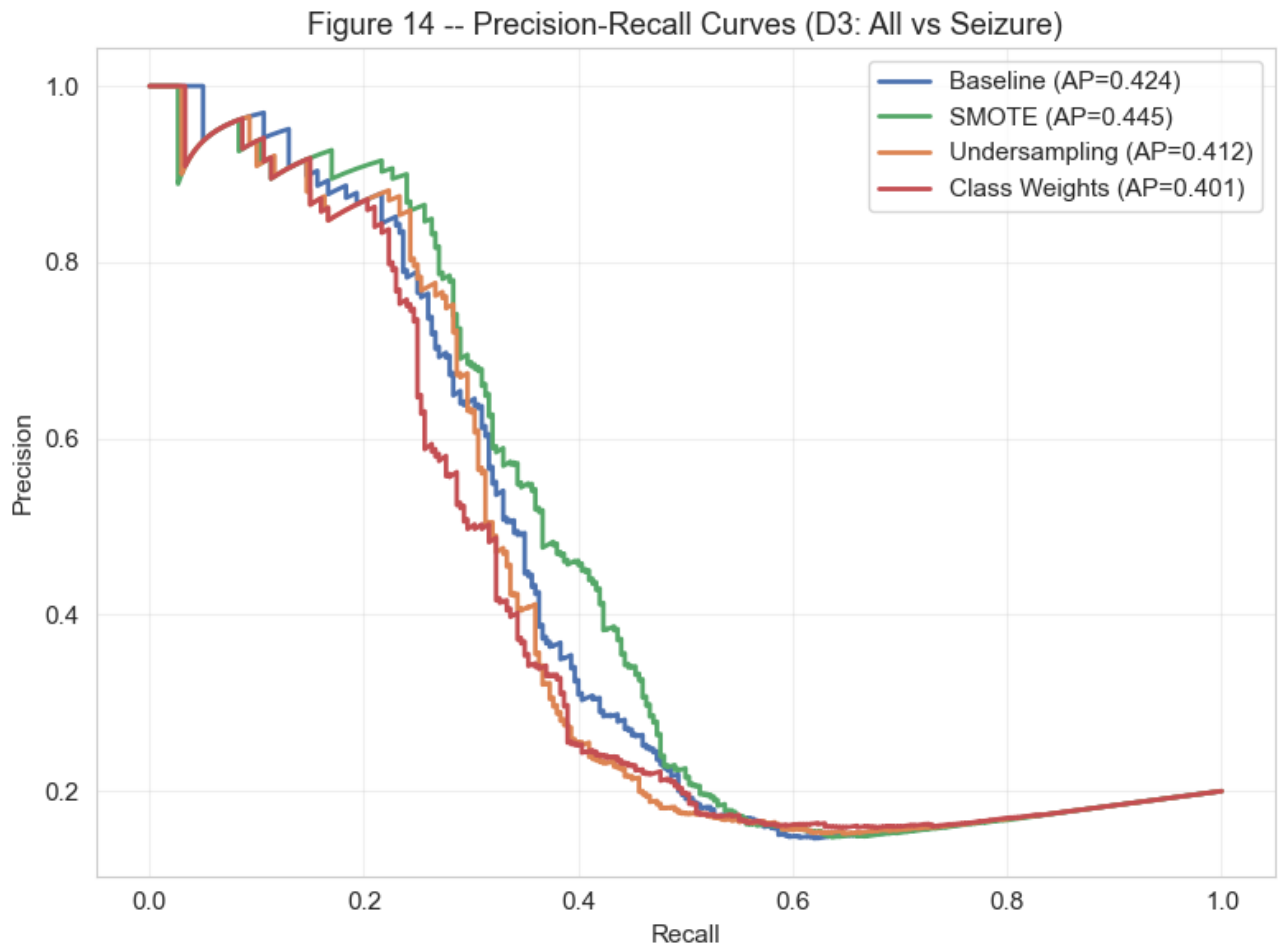


Figure 14. Precision-Recall curves Comparison

VIII. COMPARATIVE ANALYSIS

A. RQ1: Does Preprocessing Order Affect Results?

YES, the impact is dramatic and unambiguous. Pipeline B achieves F1 of 1.000, 0.925, and 0.908 on D1, D2, D3, while Pipeline A achieves only 0.503, 0.233, and 0.130 respectively. The 50-78% absolute F1 gap demonstrates that preprocessing design is the single most impactful factor in seizure classification. The key finding: EEG seizure signatures reside in the spectral domain (band powers, entropy), not raw amplitudes. Any pipeline that extracts spectral features first will substantially outperform raw signal approaches, regardless of downstream model configuration.

Figure 15. Pipeline A vs B Comparison across all Datasets

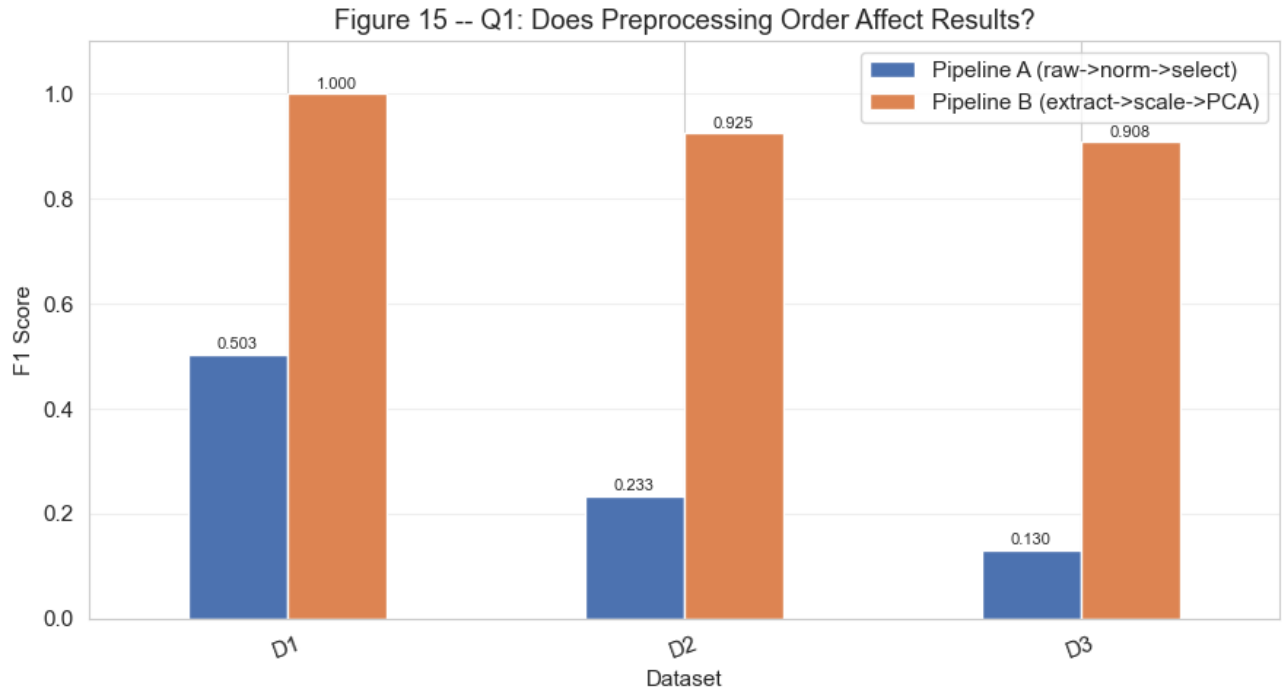


Figure 15. Pipeline A vs B Comparison across all Datasets

B. RQ2: Which Regularization Generalizes Best Across Datasets?

L1 (Lasso) achieves the highest CV F1 on all three datasets (D1: 0.9987, D2: 0.9571, D3: 0.9366). Its automatic feature selection via sparsity is beneficial even after PCA dimensionality reduction, as it identifies and weights the most discriminative principal components. L1 is recommended as the default regularization strategy for EEG-based seizure classification.

Figure 16. Best Regularization F1 per Dataset

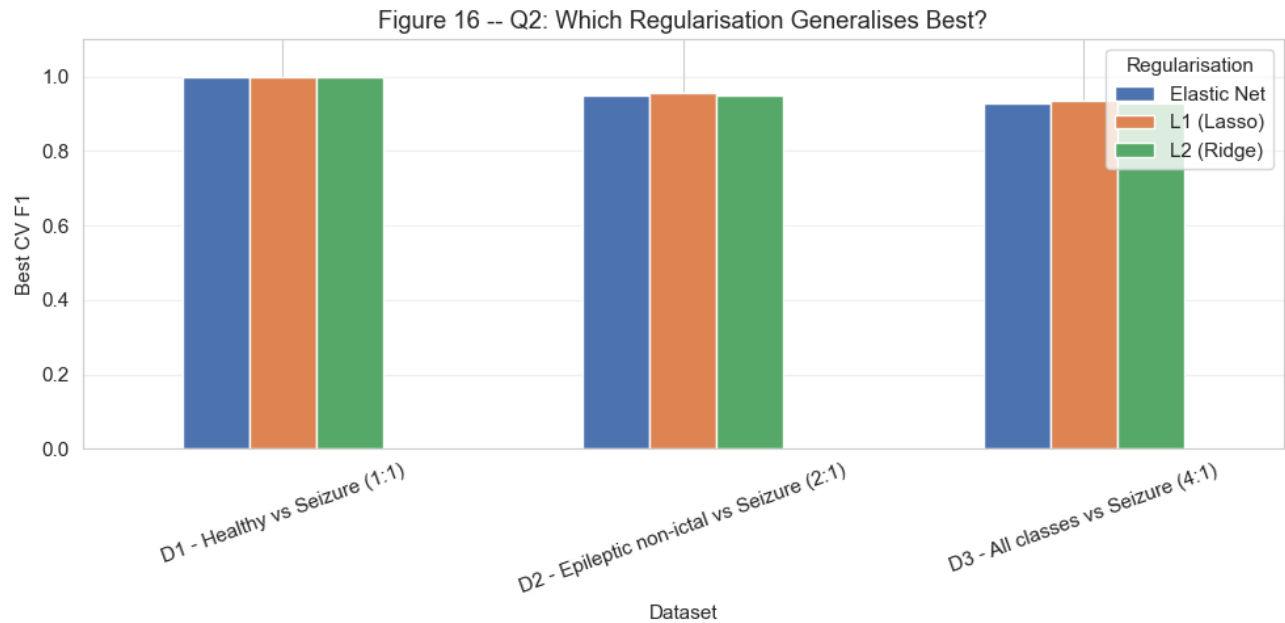


Figure 16. Best Regularization F1 per Dataset

C. RQ3: Does Elastic Net Consistently Outperform L1/L2?

NO, Elastic Net does not consistently outperform either penalty individually. On D1 and D2, L1 outperforms Elastic Net. Elastic Net slightly surpasses L2 on D3 only. The theoretical advantage of Elastic Net (exploiting correlated features via grouping) is negated by PCA orthogonalization, which removes all inter-feature correlations. This finding has practical implications: Elastic Net adds hyperparameter complexity (`l1_ratio`) without proportional performance benefit when features are orthogonalized.

Figure 17. Elastic Net vs L1/L2

Figure 17 -- Q3: Elastic Net vs L1/L2 Consistency (Spider Chart)

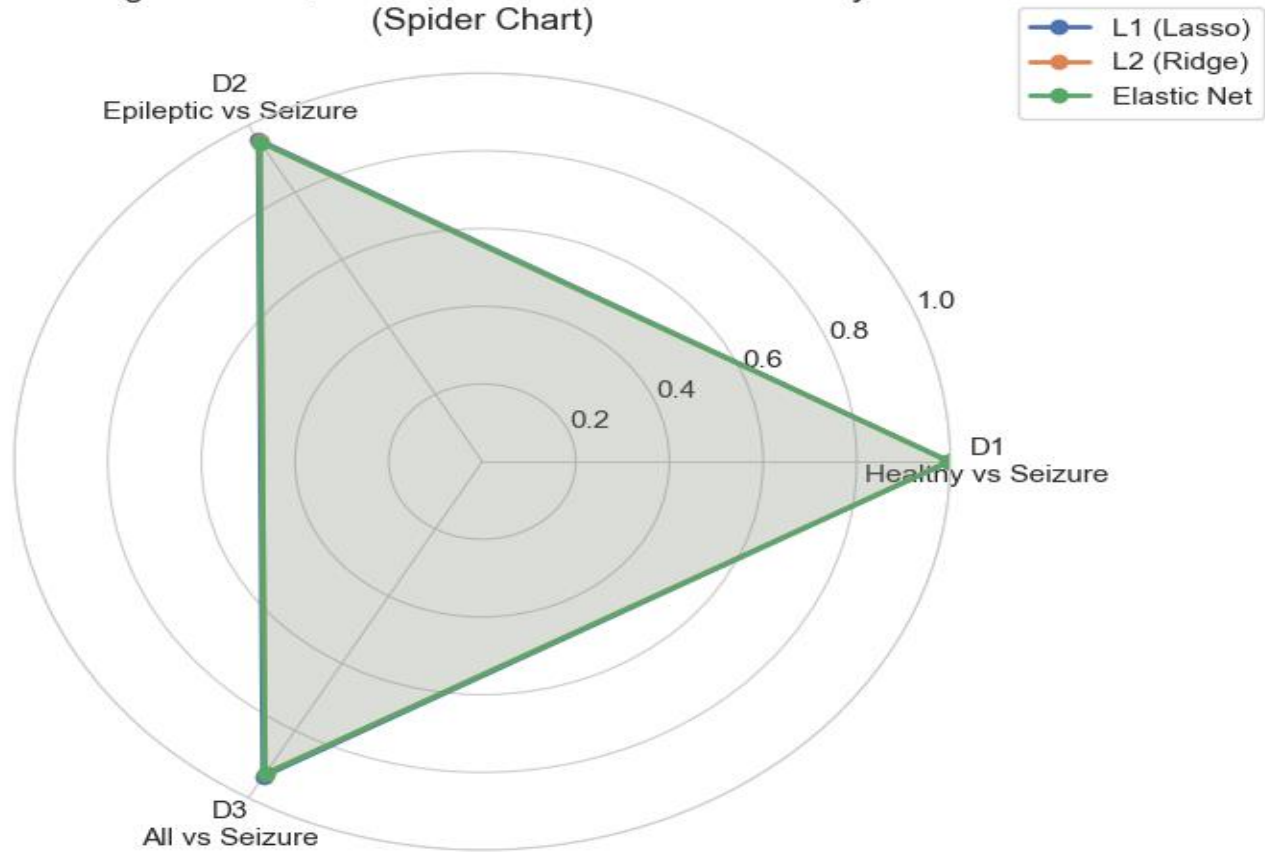


Figure 17. Elastic Net vs L1/L2

D. RQ4: How Does Imbalance Handling Interact with Regularization?

Table V. Regularization x Imbalance Interaction (D3, F1 Score)

Regularisation	SMOTE	Undersampling	Class Weights	No Resampling
L1 (Lasso)	0.927	0.934	0.927	0.923
L2 (Ridge)	0.931	0.941	0.938	0.908
Elastic Net	0.931	0.940	0.934	0.915

The interaction heatmap reveals a striking finding: L1 regularization is inherently more robust to class imbalance than L2 or Elastic Net. Without any resampling, L1 achieves $F1 = 0.923$, nearly identical to its peak with resampling (0.934). In contrast, L2 suffers a 3.3% F1 drop without resampling (0.908 vs 0.941). L1's aggressive feature zeroing effectively suppresses majority class noise, partially compensating for imbalance. For practitioners, this means L1 regularization can reduce or eliminate the need for expensive resampling steps in moderately imbalanced datasets.

Figure 18. Regularization x Imbalance Interaction Heatmap

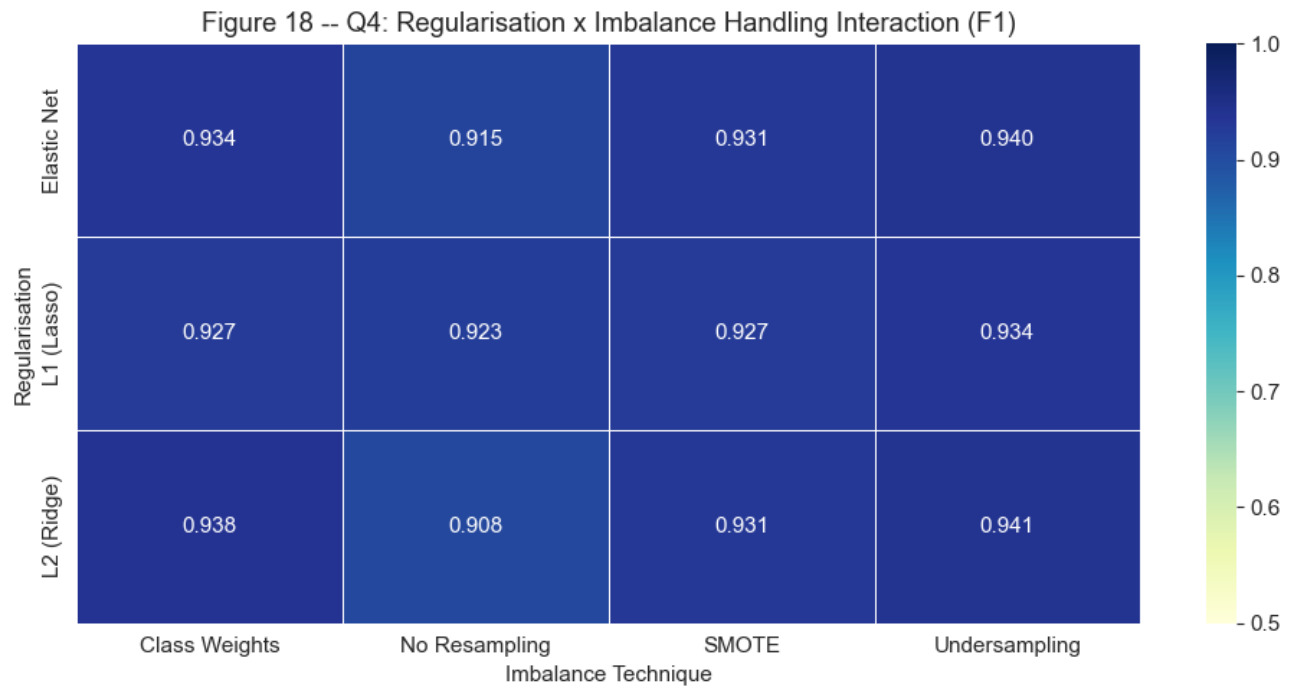


Figure 18. Regularization x Imbalance Interaction Heatmap

IX. CONCLUSIONS

This study provides a systematic investigation of how preprocessing, regularization, and imbalance handling interact in logistic regression-based EEG seizure classification. Five key conclusions are drawn:

- Preprocessing pipeline design is the most critical factor. Feature engineering (Pipeline B) yields up to 50% absolute F1 improvement over raw signal processing (Pipeline A), making it the dominant performance driver.
- L1 (Lasso) regularization provides the best generalization across all three dataset scenarios. Its sparsity property naturally selects the most discriminative features even after PCA.
- Elastic Net does NOT consistently outperform individual penalties when features are orthogonalized by PCA. Its advantage is context-dependent and disappears in this setting.
- SMOTE is the recommended imbalance handling technique for seizure prediction, as it maximizes recall (sensitivity), which is clinically paramount for avoiding missed seizure events.
- L1 regularization is inherently more robust to class imbalance than L2 or Elastic Net, with minimal F1 degradation when resampling is omitted, a practically important finding for resource-constrained clinical deployments.

REFERENCES

- [1] R.G. Andrzejak et al., "Indications of nonlinear deterministic and finite-dimensional structures in time series of brain electrical activity," *Phys. Rev. E*, vol. 64, 2001.
- [2] N.V. Chawla et al., "SMOTE: Synthetic Minority Over-sampling Technique," *J. Artif. Intell. Res.*, vol. 16, pp. 321-357, 2002.
- [3] R. Tibshirani, "Regression shrinkage and selection via the lasso," *J. Royal Stat. Soc. B*, vol. 58, pp. 267-288, 1996.
- [4] A.E. Hoerl, R.W. Kennard, "Ridge Regression: Biased Estimation for Nonorthogonal Problems," *Technometrics*, vol. 12, pp. 55-67, 1970.
- [5] H. Zou, T. Hastie, "Regularization and variable selection via the elastic net," *J. Royal Stat. Soc. B*, vol. 67, pp. 301-320, 2005.
- [6] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *JMLR*, vol. 12, pp. 2825-2830, 2011.
- [7] G. Varoquaux et al., "Assessing and tuning brain decoders," *NeuroImage*, vol. 145, pp. 246-279, 2017.